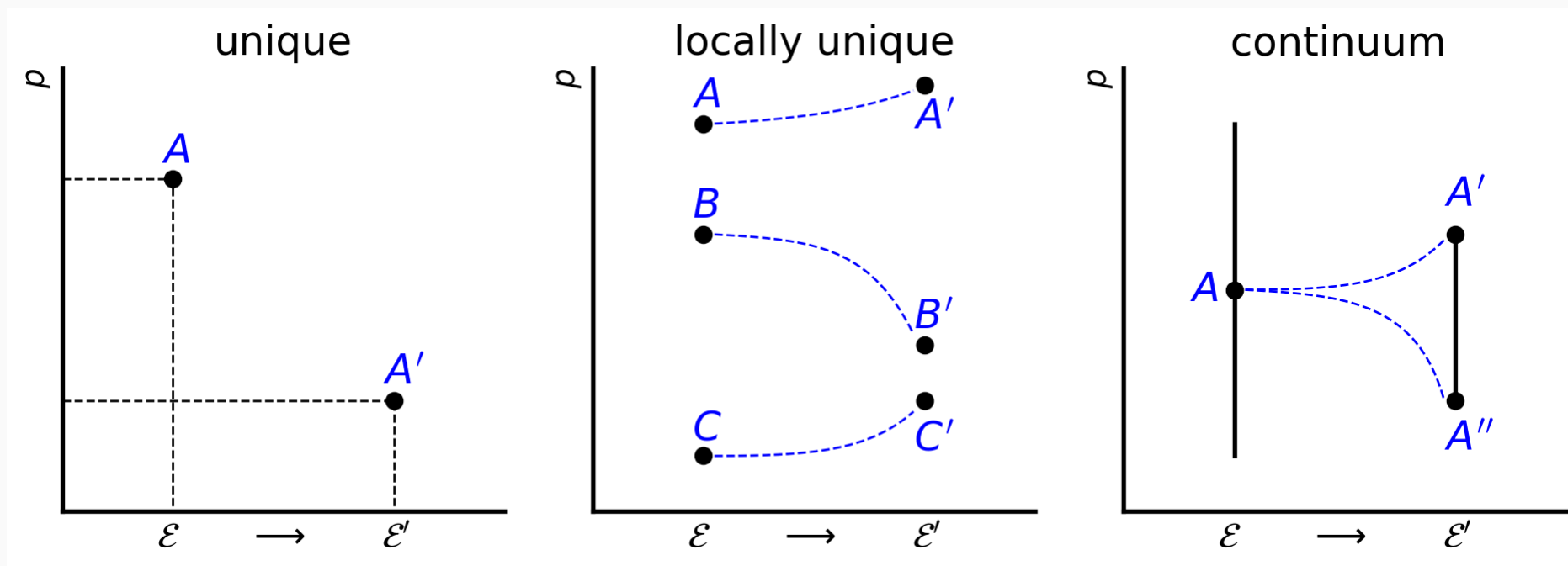


Uniqueness, Determinacy, Genericity, Stability

Uniqueness, Determinacy, Genericity

- Comparative statics (CS): How does a WE change when the underlying parameters of the economy change?
- Meaningful comparative statics in two cases,
 - Uniqueness of equilibria and
 - Local uniqueness plus a *nice transition*.



1. Uniqueness of WE and comparative statics

- If WE is unique (before and after a change in parameters), the policy change is univocally associated to a given change in the Walrasian equilibrium.
- There are no uniqueness results under reasonable assumptions on individual preferences.
- See Mas-Colell, Whinston, and Green (1995), 17.F.

2. Local uniqueness and comparative statics

Let $Eq(\mathcal{E})$ be the set of equilibrium prices of the economy $\mathcal{E}$.

2.a. Local uniqueness of WE: $\forall p^* \in Eq(\mathcal{E}), \exists \delta > 0$ such that there is no equilibrium price $q^* \in \Delta$ with $q^* \neq p^*, \|q^* - p^*\| < \delta$ and

2.b. For sufficiently small change in parameters, the number of equilibria is unchanged and each equilibrium moves nicely (i.e. continuously or differentiably) as the parameter changes.

These requirements are not sufficient for a meaningful comparative static exercise.

Main findings

1. Identification of a *class* of economies (called regular) where 2.a and 2.b hold (i.e. there are finitely many WE and the WE vary *nicely* with the parameters of the economy).
2. Genericity: “A large number of economies are within the identified class” Debreu (1970) using Sard’s Theorem.
3. Link to stability of equilibria. Index Theorem (Poincar'e-Hopf Theorem).

Preliminaries

Definition 1 Let $\mathcal{E}$ be a pure exchange economy. Agent i 's excess demand is $z : \mathbb{R}_{++}^L \times \mathbb{R}^L$,

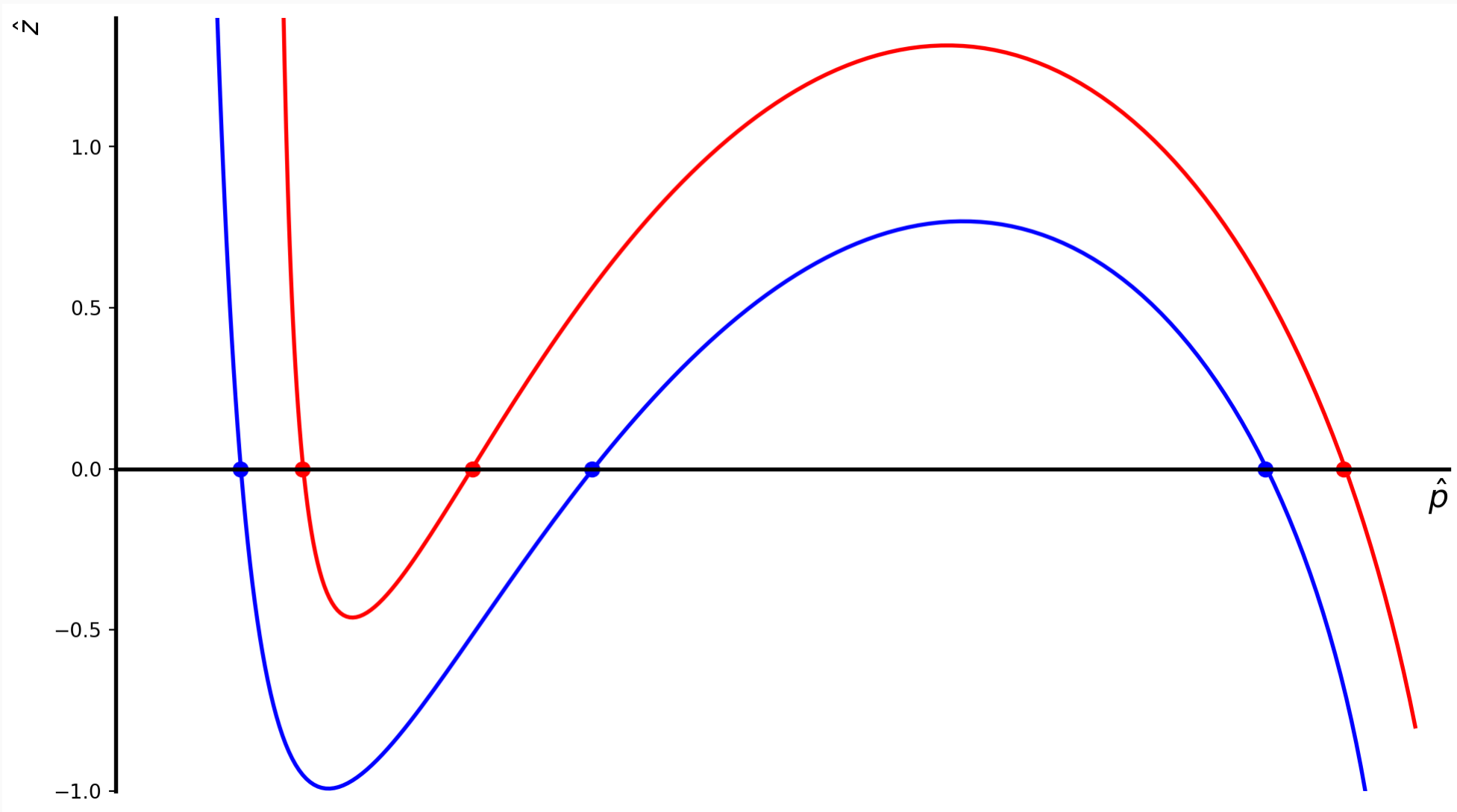
$$z_i(p, \omega_i) = D_i(p, \omega_i) - \omega_i$$

The aggregate excess demand is $z : \mathbb{R}_{++}^L \times \mathbb{R}^{LI}$,

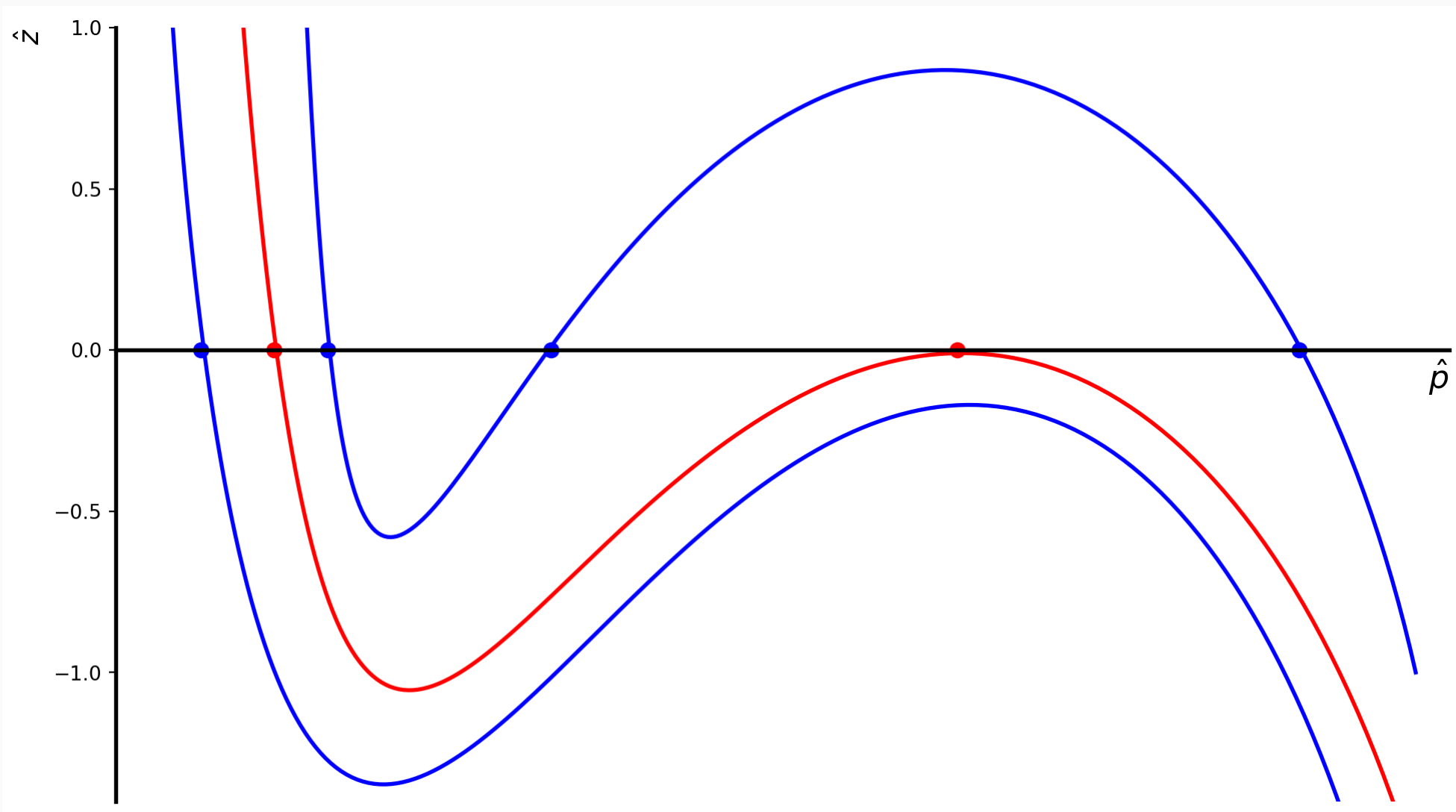
$$z(p, \omega) = \sum_{i=1}^I z_i(p, \omega_i) = \sum_{i=1}^I D_i(p, \omega_i) - \bar{\omega}$$
$$z_\ell(p, \omega) = \sum_{i=1}^I z_{i\ell}(p, \omega_i)$$

Local uniqueness and CS, example

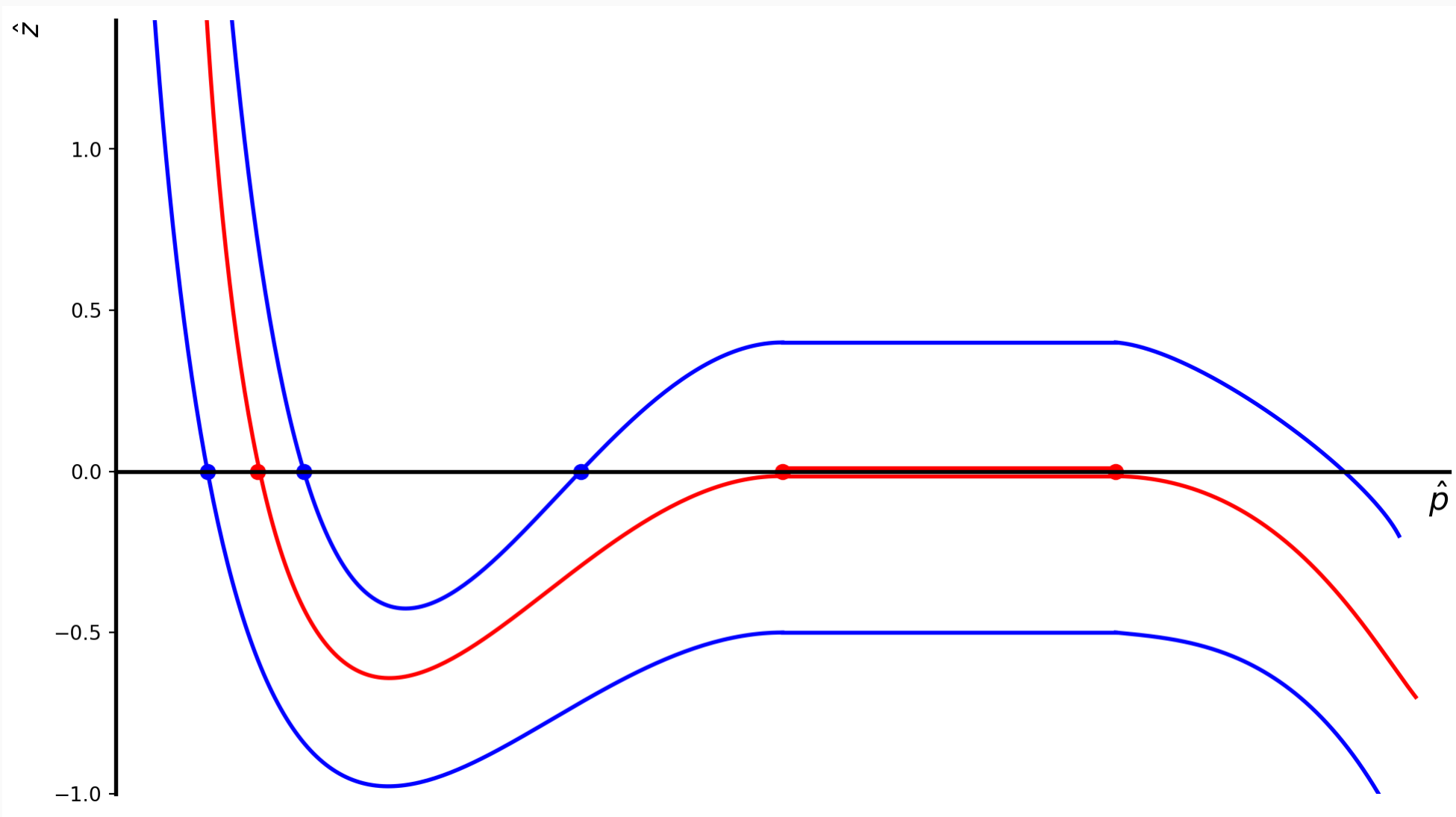
$L = 2$; Walras' law; $(\forall p \in \mathbb{R}_{++}^2) \hat{p} = \frac{p_1}{p_2}$ and $\hat{z}(\hat{p}, \omega) = z_1(\frac{p_1}{p_2}, 1, \omega)$.



Local uniqueness but



No local uniqueness



Local uniqueness–Maintained assumptions

- Let $\mathcal{E}$ be a pure exchange economy and z its aggregate excess demand.
- z satisfies the assumptions of the DGKN Lemma.
- z is homogeneous of degree zero,

$$(\forall p \in \mathbb{R}_{++}^L, \forall \alpha > 0) \quad z(\alpha p) = z(p)$$

- $z \in C^1$.

Preliminaries

- Normalize prices so that $p_L = 1$, let $\hat{p} = (p_1, \dots, p_{L-1}) \in \mathbb{R}_{++}^{L-1}$.
- Let $\hat{z} : \mathbb{R}_{++}^{L-1} \rightarrow \mathbb{R}^{L-1}$ be $\hat{z}(\hat{p}) = (z_1(\hat{p}, 1), \dots, z_{L-1}(\hat{p}, 1))$.
- [Walras' law and $p \gg 0$] $\implies (\hat{z}(\hat{p}) = 0 \iff z(\hat{p}, 1) = 0)$
- The Jacobian matrix of $\hat{z}$, $D\hat{z}$ is $L - 1 \times L - 1$.

Regularity

Definition 2 (Regular) An equilibrium price p^* of the economy $\mathcal{E}$ is regular if $\det D\hat{z}|_{\hat{p}^*} \neq 0$.

The economy $\mathcal{E}$ is regular if all its equilibrium prices p^* are regular.

Finitely many equilibria

Theorem 1 Under the maintained hypotheses, every regular economy has finitely many equilibrium prices (up to normalization).

- See Mas-Colell, Whinston, and Green (1995), Proposition 17.D.1.
- This implies that the Walrasian equilibrium prices are locally unique.

References

Debreu, Gerard. 1970. "Economies with a Finite Set of Equilibria." *Econometrica* 38: 387–92.

Mas-Colell, Andreu, Michael Whinston, and Jerry Green. 1995. *Microeconomic Theory*. New York, Oxford: Oxford University Press.